TECHNICAL PROJECT REPORT

Retail Sales Data Exploration & Analysis Dashboard

Data Analysis with Python Internship — Task 1

Project area	Retail sales data analysis
Primary language	Python
Main libraries	Pandas, NumPy, Matplotlib, OpenPyXL
Notebook	Jupyter Notebook
Data size	1,000 rows × 9 columns

Prepared as a project submission report. Replace or add your name, internship provider, and submission date if required.

1. Executive Summary

This project explores a retail sales dataset using Python. The objective is to inspect the data, verify its quality, calculate useful sales metrics, and create visual summaries that help explain revenue patterns and customer/product performance.

The dataset contains 1,000 transactions and nine columns covering transaction identifiers, dates, customer identifiers, gender, age, product category, quantity, unit price, and total amount. Data inspection found no missing values and no duplicate rows in the checked dataset. Dates were converted to a datetime format to support time-based analysis.

Key reported metrics

Metric	Result
Total revenue	456,000

Total transactions	1,000
Total items sold	2,514
Unique customers	1,000

The analysis produced monthly revenue and grouped summaries, along with charts intended to communicate trends and comparisons. Cleaned data was exported to CSV and Excel formats for reuse.

Note: Currency units and dataset provenance were not specified in the project details available for this report; revenue is therefore shown without a currency symbol.

2. Project Objectives and Scope

Objectives

• Load the retail sales CSV file into a Pandas DataFrame.

• Inspect the structure, column names, data types, and summary statistics.

• Check missing values, duplicate rows, and basic value consistency.

• Convert the transaction date field to a date/time data type.

• Calculate headline metrics for revenue, transactions, items sold, and customers.

• Explore sales by month and by available customer/product dimensions.

• Create visualizations using Matplotlib.

• Export the cleaned dataset to CSV and Excel and document the workflow.

Scope

The project focuses on descriptive analysis of the supplied dataset. It summarizes what is present in the data; it does not establish causes for sales changes, forecast future sales, or prove business relationships.

Questions considered

• What is the total revenue represented in the dataset?

• How many transactions and items are recorded?

• How does revenue change across months?

• How can revenue be compared across product categories and customer groups?

• Is the dataset sufficiently clean for the planned descriptive analysis?

Expected outputs

The working project includes a Jupyter Notebook, Python analysis script, cleaned data exports, chart images, project documentation, and this technical report. Any additional dashboard file, screenshots, or demonstration video should be checked against the final submission checklist before delivery.

3. Dataset Description

The source file used in the project is named **retail_sales_dataset.csv**. The inspected dataset has 1,000 rows and nine columns.

Column	Description
Transaction ID	Identifier for a transaction
Date	Transaction date; converted to datetime
Customer ID	Identifier for a customer
Gender	Customer gender category as supplied
Age	Customer age
Product Category	Product category recorded for the transaction
Quantity	Quantity purchased
Price per Unit	Price per unit recorded in the row
Total Amount	Total amount recorded for the transaction row

Tools and libraries

• **Python:** main programming language for analysis.

• **Pandas:** data loading, cleaning, grouping, and tabular summaries.

• **NumPy:** numerical operations and checks.

• **Matplotlib:** charts and visual summaries.

• **OpenPyXL:** support for Excel workbook output.

• **Jupyter Notebook:** interactive execution and explanation of analysis steps.

The working environment was set up on a Windows laptop with Python and the listed libraries installed. The notebook and script provide a reproducible path from data loading to output generation.

4. Data Loading and Quality Checks

Loading and inspection

The CSV file was loaded into a Pandas DataFrame. Initial inspection used the DataFrame shape, column listing, data types, sample rows, and summary statistics to understand the dataset before analysis.

Example workflow:

```
import pandas as pd
import numpy as np

df = pd.read_csv('data/retail_sales_dataset.csv')
print(df.shape) print(df.info())
print(df.isnull().sum())
print(df.duplicated().sum())
```

Cleaning and validation

- Missing-value check: no missing values were reported across the inspected columns.
- Duplicate check: no duplicate rows were reported.
- Date handling: the Date column was converted to a datetime data type.
- Basic range checks: the checks recorded no invalid values for the reviewed quantity, price, and age fields.
- Consistency check: the quantity multiplied by unit price matched Total Amount for all 1,000 rows in the checkperformed.

Why these checks matter

Missing values can affect group summaries, duplicates can inflate totals, and inconsistent numeric values can distort revenue calculations. Converting dates allows rows to be grouped by month. The checks above support the use of the dataset for the descriptive analysis carried out in this project.

Quality checks only describe the conditions tested. They do not guarantee that every business rule, identifier, or source-system issue has been verified.

5. Exploratory Data Analysis

Headline metrics

The notebook calculated four summary metrics from the inspected data:

- **Total revenue:** 456,000
- **Total transactions:** 1,000
- **Total items sold:** 2,514
- **Unique customers:** 1,000

Monthly revenue

The Date field was used to derive a month-level grouping and aggregate Total Amount. This enables a month-by-month view of revenue and supports a trend chart. Example values recorded during analysis include 46,580 for October 2023, 34,920 for November 2023, and 44,690 for December 2023. January 2024 was recorded as 1,530 in the available output.

These figures should be interpreted in the context of the supplied dataset and its coverage. A lower monthly total may reflect the dataset's date range or incomplete coverage rather than a real-world business decline.

Product and customer comparisons

Grouped summaries were prepared to compare revenue by product category, gender, and age group. Such comparisons can help describe which groups contribute different amounts of recorded revenue. The report does not assign category rankings because the exact complete grouped results were not available in the project notes used to prepare this PDF.

Reproducible aggregation examples

```
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.to_period('M')
monthly_revenue = df.groupby('Month')['Total Amount'].sum()

revenue_by_category = df.groupby('Product Category')['Total Amount'].sum()
revenue_by_gender = df.groupby('Gender')['Total Amount'].sum()
```

6. Data Visualization and Dashboard

Matplotlib was used to create visual summaries from the grouped data. The project notes identify the following chart outputs:

- **Monthly revenue trend:** illustrates how recorded revenue varies by month.
- **Revenue by product category:** compares total revenue across product categories.
- **Revenue by gender:** compares recorded revenue across gender groups.
- **Revenue by age group:** compares revenue across defined age bands.

Chart design and interpretation

A line chart is appropriate for monthly revenue because the x-axis represents time. Bar charts are appropriate for category and customer-group comparisons. Chart titles, axis labels, readable category names, and consistent units should be used to make the results clear.

The image files were saved separately in the project visualization folder during the workflow. They are not embedded in this PDF because the image files were not available in the report-generation workspace at the time of creation. Add the actual chart images here or attach them separately if the internship submission expects a single report containing the figures.

Dashboard

The dashboard is intended to present the headline metrics and charts in one place for easier review. Before submission, open the final dashboard file in its target application and confirm that visuals, labels, filters, and data sources work as expected. Do not mark a dashboard deliverable complete unless the working file is included in the submission.

Suggested checks

• Verify all chart images open correctly.

• Confirm each chart uses the cleaned dataset.

• Check that chart labels and totals match the notebook summaries.

• Ensure the dashboard and report use the same revenue definitions.

7. Exporting Results and Reproducibility

Cleaned data exports

The cleaned dataset was exported to CSV and Excel formats. The recorded output names are **retail_sales_cleaned.csv** and **retail_sales_cleaned.xlsx**. CSV provides a portable plain-text format, while Excel supports convenient review and sharing.

Project organization

Folder / file	Purpose
data/	Original dataset and cleaned exports
notebooks/retails_sales_analysis.ipynb	Interactive analysis and outputs
scripts/sales_summary.py	Reusable summary script
visualizations/	Saved chart images
reports/	Project summaries and report outputs
README.md	Setup and usage documentation

Re-running the project

• Install Python and the required libraries.

• Place the original CSV in the expected data folder.

• Open the notebook and run cells from top to bottom.

• Run the summary script from the project root if required.

• Confirm that cleaned exports and chart files are generated in the expected folders.

Use relative paths where practical and keep the input dataset unchanged so that results can be reproduced. Record any changes to the dataset or analysis code before regenerating outputs.

8. Findings, Limitations, and Conclusion

Findings supported by the analysis

• The inspected dataset contains 1,000 rows and nine columns.

• The checks reported no missing values and no duplicate rows.

• The quantity × unit price consistency check matched Total Amount in all 1,000 rows tested.

• The calculated summary metrics were total revenue 456,000, 1,000 transactions, 2,514 items sold, and 1,000unique customers.

• Monthly and grouped revenue summaries were produced to support visual exploration.

Limitations

• The currency and source provenance were not specified in the information available for this report.

• Results describe the supplied dataset and should not be generalized beyond its coverage without additionalevidence.

• No causal analysis, forecasting, or statistical significance testing was performed.

• The report-generation workspace did not contain the chart images, so they are referenced but not embeddedhere.

• Dashboard, screenshots, GitHub publication, and demonstration-video completion must be verified separately.

Conclusion

This project demonstrates a basic retail sales analysis workflow using Python: importing data, inspecting its structure, checking quality, deriving summary metrics, grouping sales by time and customer/product dimensions, visualizing results, and exporting cleaned data. The work provides a foundation for more advanced analysis and business reporting.

Recommended next steps

• Embed the saved charts into the report if required by the evaluator.

• Verify that the dashboard file opens and is included.

• Run the notebook from a clean session and confirm that all cells execute.

• Check the README, cleaned CSV/Excel files, screenshots, and source code.

• Complete the required demonstration video and repository submission.

End of report